Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
  Black: Molex 2147601225
  Red: Molex 2147602225

Reverse Polarity Protection

Q13
CSD19534Q5A

J10
POWER
V_IN
V_IN_RTN
Molex
0436500200

D1
SMDJ24CA
24V

C57
10uF-50V-10%-X7R-1210

C55
0.1uF-50V-10%-0603

U7
LM74700QDBVRQ1
VCAP
EN
GND
A
G
C

C48
10uF-50V-10%-X7R-1210

R28
5.1k-1%-0603

R30
5.1k-1%-0603

R25
5.1k-1%-0603

D2
PWR

VIN_PS

File: buck-boost-power-supply.kicad_sch
2/5
+V_IN
-V_IN
15V 3A Buck-Boost Power Supply
+V_OUT
-V_OUT

Power Supply Back
Feed Protection

Q1
CSD19534Q5A

C1
10uF-50V-10%-X7R-1210

C13
0.1uF-50V-10%-0603

U1
LM74700QDBVRQ1
VCAP
EN
GND
A
G
C

C14
10uF-50V-10%-X7R-1210

Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
  Black: Molex 2147601225
  Red: Molex 2147602225

J4
V_OUT
V_OUT_RTN
15V
Molex
0436500200

Mating Connector:
Housing: Molex 5040510401
Precrimped 24AWG wire 150mm: Molex 0797581003

J7
USB1
Molex
5040500491

File: canable2.kicad_sch
3/5
CANable2.0
USB_DP
USB_DM
3V3_USB
GND_CAN
CAN_HI
CAN_LO

J9
CAN1
MFG P/N: 0705550036
MFG: Molex

J1
Terminal

J2
Terminal
Keystone Electronics
7774

File: NX-Button-Interface.kicad_sch
4/5
NX-Button Interface

File: usb-serial.kicad_sch
5/5
USB Serial
3V3
RXD
TXD
RESET
FORCE_RECOVERY
3V3_BTN
UART2_TXD
UART2_RXD
SYS_RESET
REC
NX_PC_LED+
NX_PC_LED-

J3
NX CONTROL & UART
3V3_BTN
NX_PC_LED-
NX_PC_LED+
UART2_RXD
UART2_TXD
AC OK
Auto Power On Disable
SYS_RESET
REC
GND
SLEEP/WAKE*
Molex
PH1RB-15-UA
GND1

D3
PC

SW2
RECOVERY

SW1
RESET
GND1

H3
MountingHole
H10
MountingHole
H11
MountingHole
H4
MountingHole

H5
MountingHole
H7
MountingHole
H8
MountingHole

H1
1.152mm ToolingHole
H12
1.152mm ToolingHole
H2
1.152mm ToolingHole
H13
1.152mm ToolingHole

Light Pipe
LP1
Bivar
Bivar-SLP3-150-100-F
LP2
Bivar
Bivar-SLP3-150-100-F
LP3
Bivar
Bivar-SLP3-150-100-F
LP4
Bivar
Bivar-SLP3-150-100-F

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1868 SPACE COOKIES

Sheet: /
File: TVS-Adapter-v1.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-19	Rev: 1
KiCad E.D.A. 10.0.7		Id: 1/5

+V_IN
-V_IN
+V_OUT
-V_OUT
GND
C8 47uF 50V
C12 47uF 50V
10uF-50V-10%-X7R-1210
C47
C10
C9
C7
C11
C46
C52
C53
C6
C17
C21
C20
C16
C26
C25
C2
C4
C5
C3
HDRV1
LDRV1
Q12 CSD19534Q5A
Q11 CSD19534Q5A
Q5 CSD17578Q5A
Q8 CSD17578Q5A
HDRV2
LDRV2
SW1
SW2
L1 2.2uH-19A-5.1mohm-20%
RCS
CSG
R20 10mOhm-3W-1%-2512
D9 BAT54KFILM
R29 249K-1%-0603
R26 93.1K-1%-0603
R23 2K-1%-0603
R27 10R0-1%-0603
C41 0.1uF-50V-10%-0603
C40 0.1uF-50V-10%-0603
R5 357K-1%-0603
R9 20K-1%-0603
Texas Instruments
LM5176PWPR
U4
VIN
VISNS
EN/UVLO
SS
COMP
MODE
RT
DITH
SLOPE
CS
CSG
VCC
BIAS
VOSNS
HDRV1
BOOT1
SW1
LDRV1
FB
PGOOD
HDRV2
BOOT2
SW2
LDRV2
ISNS(+)
ISNS(-)
AGND
PGND
EP
Vcc
D8 BAT54KFILM
C39 1uF-50V-10%-0603
C42 0.1uF-50V-10%-0603
D6 BAT54KFILM
C31 0.1uF-50V-10%-0603
R31 0R-1%-0603
R24 0R-1%-0603
R16 0R-1%-0603
R19 0R-1%-0603
15nF-10%-50V-0603 C35
5.6nF-10%-50V-0603 C28
5.6nF-10%-50V-0603 C30
270pF-10%-50V-0603 C33
5.9K-1%-0603 R10
205K-1%-0603 R22
13K-1%-0603 R21
150pF-50V-10%-0603 C37
C29 47pF-50V-10%-0603
100R-1%-0603 R12
100R-1%-0603 R11

VIN(MAX)	24V
VIN(MIN)	5V
VOUT	15V
IOUT(MAX)	3A
FSW	589 kHz
Efficiency	96.7%

15V-3A-Power Supply Analysis

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Sheet: /15V 3A Buck-Boost Power Supply/
File: buck-boost-power-supply.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-19	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 2/5

5V-250mA-Power Supply Analysis

Reverse Polarity Protection

Q2
SI2301CDS-T1-GE3

3V3_USB

3V3

L2
2.2uH-41mOhm-20%

U2
TPS613222ADBV
SW
VOUT
GND

5V_CAN

C15
22uF-10%-16V-0805

C18
22uF-10%-16V-0805

C19
22uF-10%-16V-0805

GND_CAN

Q10
SI2301CDS-T1-GE3

3V3_CAN

R18
93.1k-1%-0603

R13
1M-1%-0603

D7
BAT54KFILM

Q9
SS8050

C32
22uF-10%-16V-0805

3V3_CAN delays bringing CAN RX (BOOT0) high until NPN turns on. ~10sec.

3V3

LQFP

R6
13K-1%-0603

C27
0.1uF-50V-10%-0603

WORD_LED

USB_DM

USB_DP

SWDIO

SWCLK

STAT_LED

NRST

CAN_RX

CAN_TX

BOOT0

R17
13K-1%-0603

STM32G431CBTx

U3

5V_CAN

C56
0.1uF-50V-10%-0603

U8
TJA1051T/3/1J
TXD
RXD
VIO
S
VCC
GND
CANH
CANL

CAN_TX

CAN_RX

3V3_CAN

C50
0.1uF-50V-10%-0603

CAN_H

CAN_L

D10
PESD1CAN,215

GND_CAN

D5
WORD

D4
STAT

R4
1K-1%-0603

R3
1K-1%-0603

WORD_LED

STAT_LED

J8
SWD
PH1RB-05-UA
Adam Tech

C24
0.1uF-50V-10%-0603

C22
0.1uF-50V-10%-0603

C23
0.1uF-50V-10%-0603

C36
0.1uF-50V-10%-0603

C34
0.1uF-50V-10%-0603

CANable 2.0 Schematic

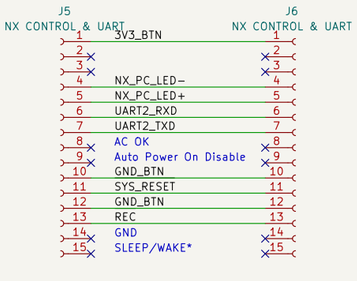
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Sheet: /CANable2.0/
File: canable2.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-19	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 3/5



H6
M3.7 MountingHole

H9
M3.7 MountingHole

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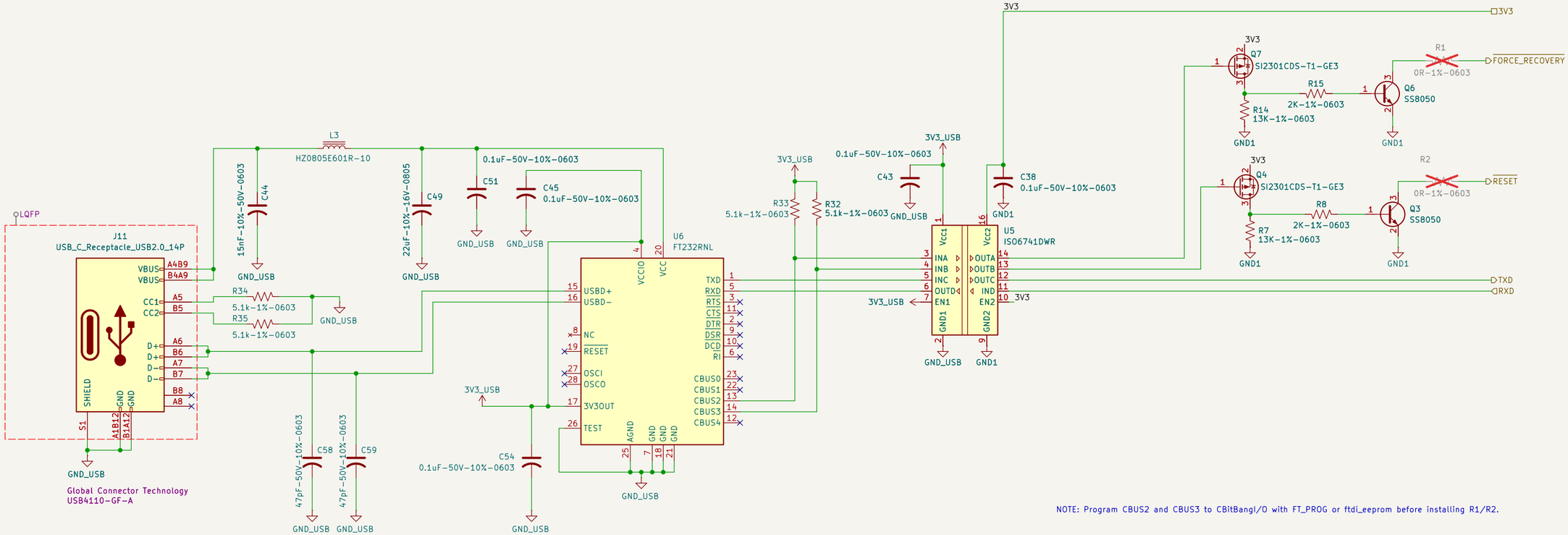
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1868 SPACE COOKIES

Sheet: /NX-Button Interface/
File: NX-Button-Interface.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-19	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 4/5



NOTE: Program CBUS2 and CBUS3 to CBitBangI/O with FT_PROG or ftdi_eeprom before installing R1/R2.

https://hackaday.com/2023/08/07/all-about-usb-c-example-circuits/

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1868 SPACE COOKIES

Sheet: /USB Serial/
File: usb-serial.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-19	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 5/5